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Enhanced ionic liquid mobility induced by
confinement in 1D CNT membranes†Q. Berrod,^a F. Ferdeghini,^a P. Judeinstein,^a N. Genevaz,^a R. Ramos,^b A. Fournier,^b
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Water confined within carbon nanotubes (CNT) exhibits tremendous enhanced transport properties. Here, we extend this result to ionic liquids (IL) confined in vertically aligned CNT membranes. Under confinement, the IL self-diffusion coefficient is increased by a factor 3 compared to its bulk reference. This could lead to high power battery separators.

Confinement of fluids in nanopores is well known to modify the physical properties observed in bulk.¹ For instance, a typical signature of the nano-confinement of fluids is depicted by the so-called Gibbs–Thomson effect: the melting temperature downshift is inversely proportional to the pore size of the confining material.² While this effect can be blurred by non-negligible surface effects, tuning the physico-chemical properties of matter at the nanoscale has been of major interest for fundamental research for decades. Beyond the understanding of specific physical behavior, the modification of the physico-chemical properties has raised huge interest for technological developments. Nevertheless, while the chemical and/or physical problems are often of fundamental nature at the nanometer scale, the properties that should be controlled are of technological nature and lie at the macroscopic levels. For instance, H₂ storage in porous reservoirs, CO₂ sequestration, nuclear waste management, or ionic conduction in battery separators all deal with the general issue of macroscopic transport through porous structures. But, these are situations where the interplay of (i) topology, (ii) dimensionality, and (iii) competition between surface and volume interactions, pushes to its limits our understanding of the connection between the transport properties at the microscopic and macroscopic levels.

Confinement of fluids within carbon nanotubes (CNT) is particularly interesting due to their very peculiar and unusual

surface. Indeed, CNT allow very strong confinement potential (they can reach less than 1 nm in diameter) and possess frictionless or low friction surfaces.³ This unique porous system offers therefore the possibility to decouple confinement effects from surface interactions.

Recent technological developments have enabled the design of a new, appealing material: a macroscopic 1D nanoporous CNT based matrix, so-called Vertically Aligned CNT (VA-CNT).⁴ These VA-CNT have opened a path to high impact functional materials in terms of technological developments (membranes for filtration, electrochemistry, electronics *etc.*),^{5–7} as well as fundamental research: this exceptional model system with (i) no tortuosity and (ii) no or weak interaction of the confined fluid with the pore surface could have a direct relevance to the understanding of the physics from the nanoscale to the macroscopic level.

Water confined inside a 1.4 nm diameter CNT is a perfect example of the unexpected effect induced by nanometric confinement. Under such a strong confinement, water molecules adopt a very specific and frustrated organization (modification of bulk water H-bond network).⁸ As a consequence, at the macroscopic scale, when forced inside a CNT membrane, water is found to flow up to three orders of magnitude faster than predicted by the continuum hydrodynamics picture.⁹

Based on this exciting framework, we propose to extend the results obtained on water to the case of electrolytes confined in VA-CNT membranes to enhance their ionic conductivity. The goal is, in particular, to design a new concept of battery separators with high power density. This target will be achieved by combining (i) a reduction of the dimensionality (3D motions in bulk electrolytes towards 1D-like motions in VA-CNT) and (ii) a modification of the electrolyte's physico-chemical properties at the nanoscale (induced by confinement within CNT).

Ionic liquids (IL) are very good candidates to be used as electrolytes: they show a high ionic conductivity, solvate many salts (based on Li⁺, Na⁺ *etc.*), and have a low volatility and flammability (relevant for safety issues). IL have been successfully confined in CNT, and recent molecular dynamics (MD)

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simulation results show a spontaneous filling of the CNT, along with an increased diffusivity compared to bulk electrolyte.^{10–12} Moreover, ionic conductivity measurements of an IL confined in a single pore shows, at room temperature, a tenfold increase of the confined fluid macroscopic transport properties.¹³ IL are therefore promising ionic conducting electrolytes for confinement in VA-CNT membranes.

Herein, we present the preparation and characterization of the dynamics of IL (1-methyl-3-octyl-imidazolium-tetrafluoroborate (OmimBF₄) and 1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide (BmimTFSI)) confined in VA-CNT. The CNT forests (multiwall CNT with 2 or 3 walls, 4 nm internal diameter, 3×10^{11} tips cm⁻²) (Fig. 1a, c and S1†) are grown-up from a basal silicon wafer, and their interior is not clogged by “bamboo” graphene foils or any metallic catalysts (CNT forests are synthesized according to the base growth mechanism, leaving the catalyst on the substrate)¹⁴. They are transformed into membranes by (i) filling the inter-tube volume with polymer solution (here polystyrene, 20 wt% in toluene, but other polymers can be chosen depending on the targeted mechanical properties), and (ii) opening the tubes by mechanical abrasion after evaporating the solvent (Fig. 1b, d and S2†). The VA-CNT membranes are then immersed in the bulk IL and the interior of the CNT is spontaneously filled by capillarity.

The IL self-diffusion coefficient, D_s , of both IL, has been probed along the CNT axis by pulsed field gradient (PFG) NMR at the micrometer scale (details are given in the ESI†). The measurements have been performed at 25 °C and 55 °C using a spin echo sequence, and ¹⁹F and ¹H probes were used to determine the dynamics of anions and cations respectively. D_s was determined according to the equation:¹⁵

$$A(g)/A(0) = \exp(-\gamma^2 g^2 \delta^2 D_s (\Delta - \delta/3))$$

where A , γ , g , δ and Δ stand for the signal areas, the gyromagnetic ratio of the investigated nucleus, the intensity of the

magnetic field gradient, the duration of the gradient pulse and the diffusion time respectively. Deuterated polystyrene has been used to avoid parasitic contributions.

The decay of OmimBF₄ NMR signals (bulk and confined IL in the VA-CNT membranes) *versus* the gradient intensity is shown in Fig. 2. A faster decrease of the signal is clearly observed when the IL is confined within the CNT (labelled as OmimBF₄@CNT), revealing a higher diffusion coefficient. The self-diffusion coefficient of OmimBF₄ and BmimTFSI obtained from ¹⁹F and ¹H PFG-NMR measurements are given in Table 1.

To go further, the local dynamics of the OmimBF₄ cation has been probed by quasielastic neutron scattering (QENS) at the molecular scale (details are given in the ESI†). This technique offers, in particular, the advantage to probe the motions along or perpendicular to the tubes axis (inset Fig. 3 and S3–S5†). The QENS spectra measured in both directions are shown in Fig. 3 and compared to the bulk reference.

Local diffusion coefficients can be extracted from QENS analysis (in the time and space range of [1–10 ps] and [Å–nm] respectively). However, in this study, the broadening of the quasielastic signal over the whole Q-range of bulk IL and confined IL in both orientations, is very similar (Fig. 3, S4 and

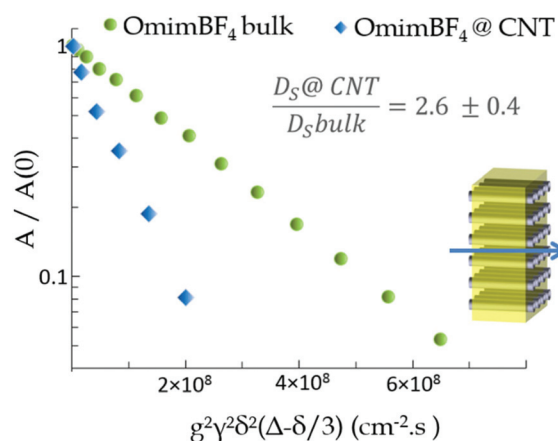


Fig. 2 ¹⁹F NMR signal decay of bulk OmimBF₄ (green) and OmimBF₄ confined in the VA-CNT membrane (blue) as a function of the gradient field (¹⁹F PFG-NMR measurements, 25 °C). The faster NMR signal decay of the confined IL shows an increase of D_s (along the CNT axis).

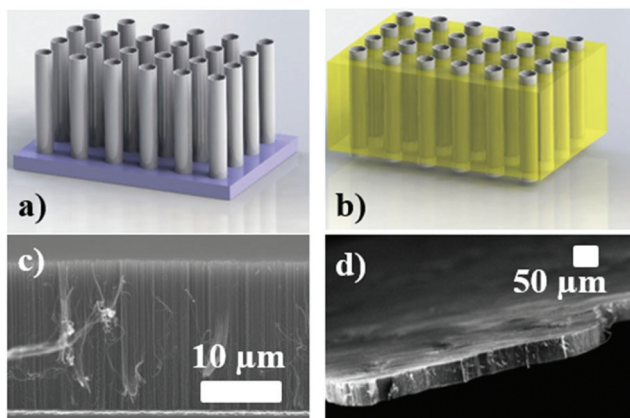


Fig. 1 Schematic representation of (a) the initial CNT forest and (b) the final CNT membrane with opened CNT on both sides. SEM images of (c) the CNT forest and (d) the CNT membrane. The membrane thickness can be tuned from 10 to 200 μm.

Table 1 ¹H (cation) and ¹⁹F (anion) self-diffusion coefficients (D_s) measured by PFG-NMR at 25 °C and 55 °C, for OmimBF₄ and BmimTFSI. The D_s ratio between IL under confinement and in bulk is given

Sample	¹ H – 25 °C (10 ⁸ cm ² s ⁻¹)	¹⁹ F – 25 °C (10 ⁸ cm ² s ⁻¹)	¹ H – 55 °C (10 ⁸ cm ² s ⁻¹)
OmimBF ₄	3.7 ± 0.2	4.7 ± 0.3	14.5 ± 1.0
OmimBF ₄ @CNT	8.2 ± 0.8	11.5 ± 1.4	24 ± 3
Ratio	2.2 ± 0.3	2.6 ± 0.4	1.7 ± 0.3
BmimTFSI	27 ± 2	21.5 ± 2	—
BmimTFSI@CNT	31 ± 3	22 ± 2	—
Ratio	1.1 ± 0.2	1.0 ± 0.1	—

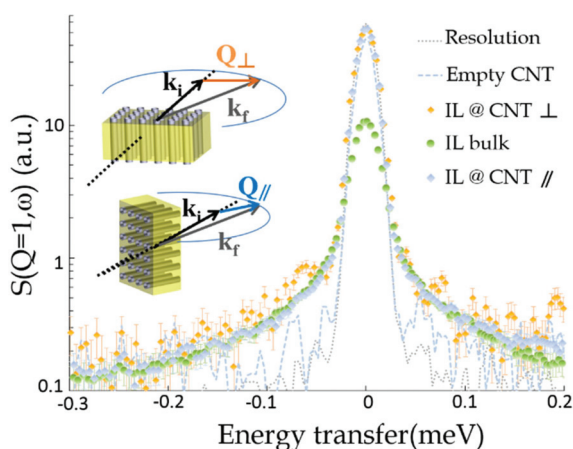


Fig. 3 $S(Q, \omega)$ spectra of bulk OmimBF₄ (green), and confined OmimBF₄ in the CNT membrane at $Q = 1 \text{ \AA}^{-1}$ (298 K). The sample has been selectively oriented to probe the dynamics along (blue) or perpendicular (orange) to the CNT (details are given in the ESI†). The gray dotted line is the resolution function (full width at half maximum: 20 μeV), and the blue dashed line the signal of the empty CNT membrane (no quasielastic contribution is observed).

S5†). It is therefore sufficient to reveal identical, diffusional and reorientational mechanisms (the higher elastic intensity, centered on 0 meV, observed for the CNT membrane arises from the static polymer matrix atoms). In a sophisticated QENS analysis, Ferdeghini found extremely fast local diffusion coefficients in bulk IL ($D_{\text{OmimBF}_4}^{\text{QENS}} = 2.9 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$, $D_{\text{BmimTFSI}}^{\text{QENS}} = 4.8 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$).¹⁶ We can consider from the QENS measurements performed here, that under confinement those values are unchanged.

Under confinement, D_s of OmimBF₄ increases, whereas no modification of the IL dynamics is measured by QENS. Both results are not in contradiction. Indeed, bulk IL diffusion coefficients obtained from QENS experiments at the molecular scale are roughly 10^2 to 10^3 times higher than the ones obtained by PFG-NMR at the micrometer scale. Measuring various diffusion coefficients depending on the probed space scale, is typically found in systems with tortuosity and/or self-organized fluids.¹⁷ It has been shown that IL tend to self-organize in transient nanodomains which certainly induce a barrier to the micrometer diffusion.^{18,19} Therefore, the self-diffusion coefficient increase observed (by PFG-NMR) for OmimBF₄ under confinement, is likely due to a modification of its nano-segregation, and do not imply a modification of the dynamics below the IL transient nanodomain size. This explanation is consistent with the results obtained for BmimTFSI (the self-diffusion coefficients do not change under confinement), which has a much less pronounced self-organization than OmimBF₄. Moreover, measuring the same diffusion coefficient in bulk and under confinement is already an interesting result showing no, or weak, interactions between the surface of the CNT and the IL. It must be noted that here, the CNT internal diameter is still quite large compared to the IL size. By decreasing the diameter down to 1–2 nm, we think the

transport property gain could reach an order of magnitude.¹³ This hypothesis is consistent with recent MD simulations: when confined in CNT, the IL mobility (probed at the nanometer scale) is bulk-like for the larger CNT (3 nm in diameter) and enhanced by a factor 5 for the smaller diameter CNT (1.4 nm).¹¹ Ohba *et al.* theoretically showed a mobility increase by a factor 2 or more than 10 when the IL is confined in a 3 or 1.4 nm diameter CNT, respectively.¹² The authors also observe a different dynamical behavior for the anion and the cation which has not been seen here experimentally (the same mobility increase has been measured for both anion and cation).²⁰ However, they studied a different IL and CNT diameters. We believe that the type of IL (chain length, chemical nature, nano-organization) and the CNT diameters are both key parameters to alter the IL mobility. Additionally, the mobility gain observed by PFG-NMR should be exalted at lower temperatures, at least for crystalline electrolytes, where they are expected to remain in the liquid state for temperatures below their melting point (Gibbs–Thomson effect).

Conclusions

To conclude, a simple way to process a 1D confining matrix with tunable mechanical properties has been shown. This appealing system possesses a very high potential in terms of technological development as well as fundamental research. Indeed, the 1D macroscopically oriented low surface friction nanotubes are an ideal system for a dynamical multi-scale analysis of confined fluids. It could bring new insights into the connection between the dynamical properties at the molecular level and the transport properties at the macroscopic scale. From a technological point of view, this material is an excellent candidate for a battery separator: it possesses tunable mechanical properties, 1D conduction paths, and high electrolyte mobility. We show here experimentally, that the IL mobility is increased by a factor 2–3 when confined in a 4 nm CNT, certainly originating from a frustration of the IL self-organization at the nanoscale. We expect this effect to be enhanced while reducing the CNT diameter. The confinement of the electrolytes in VA-CNT membranes also offers the possibility to develop new batteries operating at lower temperatures, and improves safety issues. It is noted that the pore density is currently too low and that the CNT are electrical conductors. However, it has been shown that higher density CNT forests can be synthesized,²¹ and/or CNT carpets can be mechanically densified,^{22,23} and several strategies could be employed to ensure a good electrical insulation. Although there are still many technological challenges to reach practical applications of CNT based separators, this study opens a new path, through a physical breakthrough, in achieving operational materials with strongly increased ionic conductivity.

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Author contributions

QB prepared the IL charged CNT membranes, analyzed the data and wrote the paper. QB, FF, PJ, SR, JO and JMZ performed the QENS experiment at ILL. QB, DY and RAM performed the QENS experiment at ANSTO. QB and PJ performed PFG-NMR measurements. NG, FF and JMZ delineated the conditions for the CNT membrane preparation. AF and JD synthesised the CNT forest. RR characterized the CNT forest. JMZ initiated and supervised the project. All the authors contributed to the preparation of the manuscript.

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